



Portuguese Navy Co-Development Programme

Vantarís ARC – Sovereign-grade autonomy for next-generation
maritime operations

From validated UK IP to NATO-aligned deployment

Portuguese Navy Co-Development Programme

Vantar is has been selected by the Portuguese Navy and Ministry of Defence for a multi-phase co-development programme delivering autonomous / unmanned maritime capability.

The programme advances from an 8-9 m multi-mission vessel (Phase 1, 2026 - 2027) to the 12 - 18m Xiren platform (Phase 2, 2027-2028), integrating the Vantar is ARC System-on-Chip autonomy core with NATO-compatible weapons, optics, and training suites.

Vantar is' co-development programme with the Portuguese Navy is central to its commercial and investment thesis. Rather than developing technology in isolation, Vantar is is building and validating its system inside a live naval operational framework. Built with the navy, not for the navy.

Vantar is is executing a formal, multi-phase co-development programme with the Portuguese Navy, in which the Navy defines operational requirements, participates directly in system design, supervises sea trials, and evaluates performance against real naval Concepts of Operations (CONOPS). This programme culminates in NATO-facing demonstration at REPMUS 2026 and materially removes core product, integration, and operational risk prior to wider procurement engagement.

The Expertise



Mastery of Software Architecture



Mastery of Software Architecture



Experts in Electronic Engineering



\$7m+ in R&D

The Vantaris team brings together specialists from defence, electronics, naval architecture, and advanced computing. Successful delivery of autonomy systems evaluated by national defence customers.

Collectively responsible for more than a decade of autonomy research, the team now enters into the Portuguese Navy's national USV development programme, supported by:

- Versatile Marine - 3D-printed vessel manufacture (Phase 1 platform)
- CEAD - 3D printing solutions and regional partnership
- Xiren UK - phase 2 multi-mission vessel supplier
- MSI Defence - weapons systems (FOC)
- Current Corps - optics & ISR software
- Junus Defence - CONOPS & training architecture

This collaboration delivers a full operational ecosystem - design, manufacture, autonomy, weapons, optics, and simulation - validated under the Portuguese MoD.

50

Years Experience in the Naval, Defence and Securities Sectors

35

Years Experience in Unmanned & Autonomous Technologies

Proven Technology - Multi Domain Autonomy

Vantaris builds upon the legacy of AS2, whose earlier-generation command and control systems were successfully integrated into government agency platforms across maritime and land domains. During its prior iteration, the technology was validated through national scale defence programmes, including participation in a \$500M program award for a major UAE defence contractor involving 45 operational PoCs – demonstrating exceptional performance and strategic alignment, though the contract was ultimately not advanced to delivery. Vantaris represents the next evolution of a proven control system lineage that has already demonstrated effectiveness at national scale.

Collectively, the Vantaris team has been working with Unmanned systems since 2011. Over the course of the last 14 years, the team has developed several generations of Unmanned Surface Vessels and vehicles ranging from a 3m jet-ski platform through to a weaponised 11m fast interceptor craft for littoral operations.



The Problem

Existing autonomy frameworks are built on layered operating systems designed for civilian robotics, not sovereign command environments.

- Each additional OS or middleware layer increases latency and vulnerability.
- Closed or proprietary stacks limit interoperability and vendor choice.
- Cloud-dependent control loops compromise mission security and reliability.

True autonomy requires hardware-level execution, instantaneous response, and supplier independence.



Hackers can bring ships and planes to a grinding halt. And it could become much more common

Hackers can bring ships and planes to a grinding halt. And it could become much more common

Cyber-attacks: how hackers are targeting seafarers

Ship hacked to burn US Military Oil Tanker into a Fireball

By Naveen Goud [Join Cybersecurity Insiders]

Maritime security: 'Hacking a ship is just like hacking a Tesla but bigger'

Ethical hacker says ships are wide open to cyber attack

Hackers can get in through a wireless keyboard, an unguarded printer, or even by reading a device's user manual to exploit weaknesses

Cyberattacks on shipping up 400-500% in five years, Lloyds List Intelligence

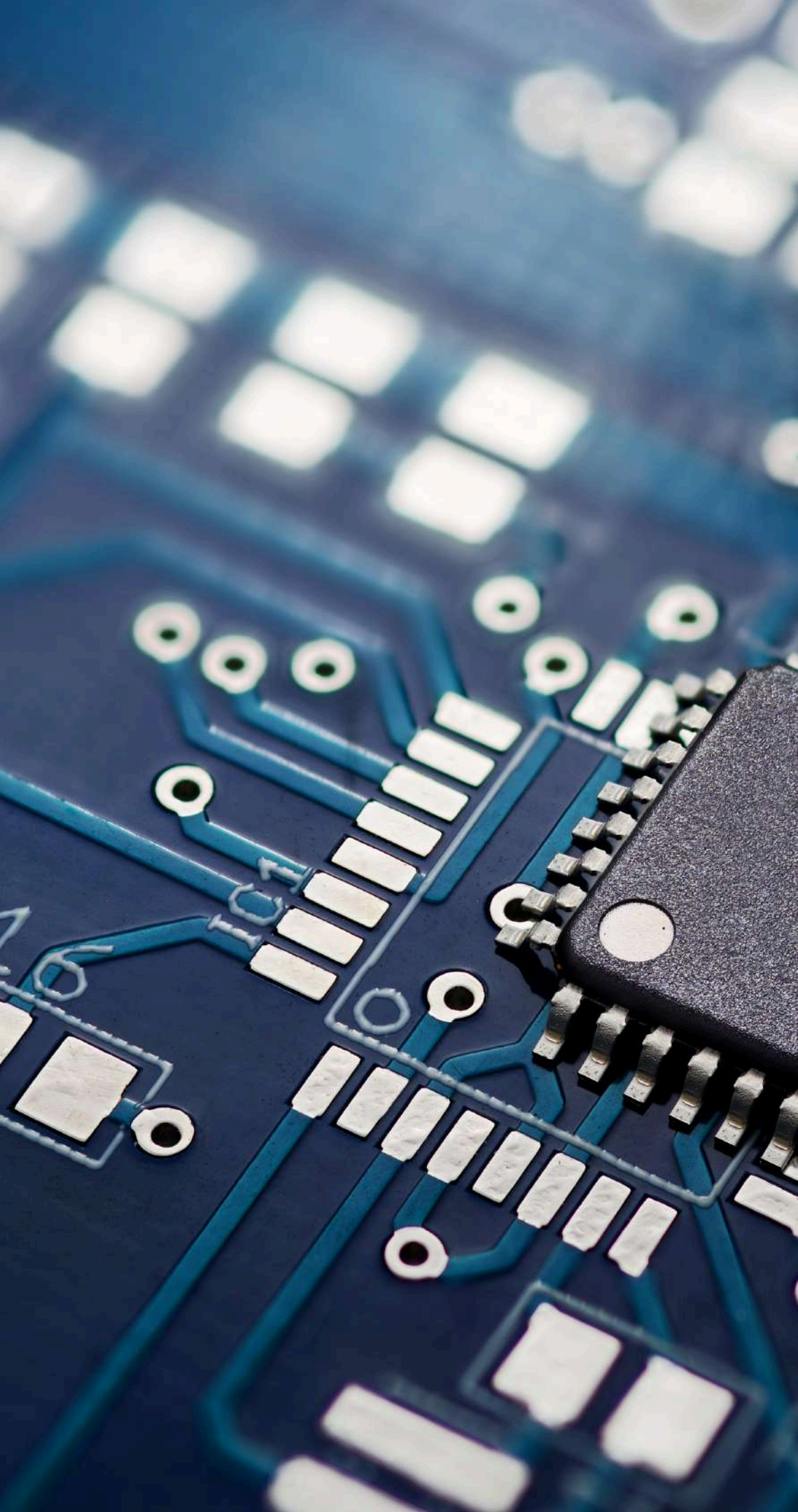
One ship is hacked every day on average

Ships and shipping companies are regularly targeted by cyber criminals, webinar discussion hears

Researcher: Not Hard for a Hacker to Capsize a Ship at Sea

In a separate and alarming incident, reports have emerged about a Portuguese cargo ship, the MV Solong, whose GPS system was reportedly hacked, causing it to stray from its designated route. The ship then collided with a stationary U.S. military oil tanker, triggering a massive explosion.

Ransomware attack on maritime software impacts 1,000 ships



Solution: Vantaris System

ARC (Autonomous Remote Command) is a proprietary bare-metal System-on-Chip (SoC) developed specifically for unmanned surface and ground vehicles, with architecture adaptable for multi-domain platforms.

Unlike conventional autonomy frameworks built atop layered or open-source operating systems, ARC runs entirely at the hardware level, free from external OS dependencies or tertiary control environments – delivering:

- Sub-0.1 s latency – 30× faster than OS-based frameworks
- Instant boot (<0.2 s) – zero start-up lag in mission hand-off
- Hardware-level security – no root access, no exploitable surface
- Fleet-Wide interoperability
- Seamless integration with 3rd party sensors, payloads, domestic tech
- Modular, smart architecture
- On-chip AI co-processor for real-time sensor fusion (EO/IR, Radar, EW)
- Seamless manned ↔ unmanned switching

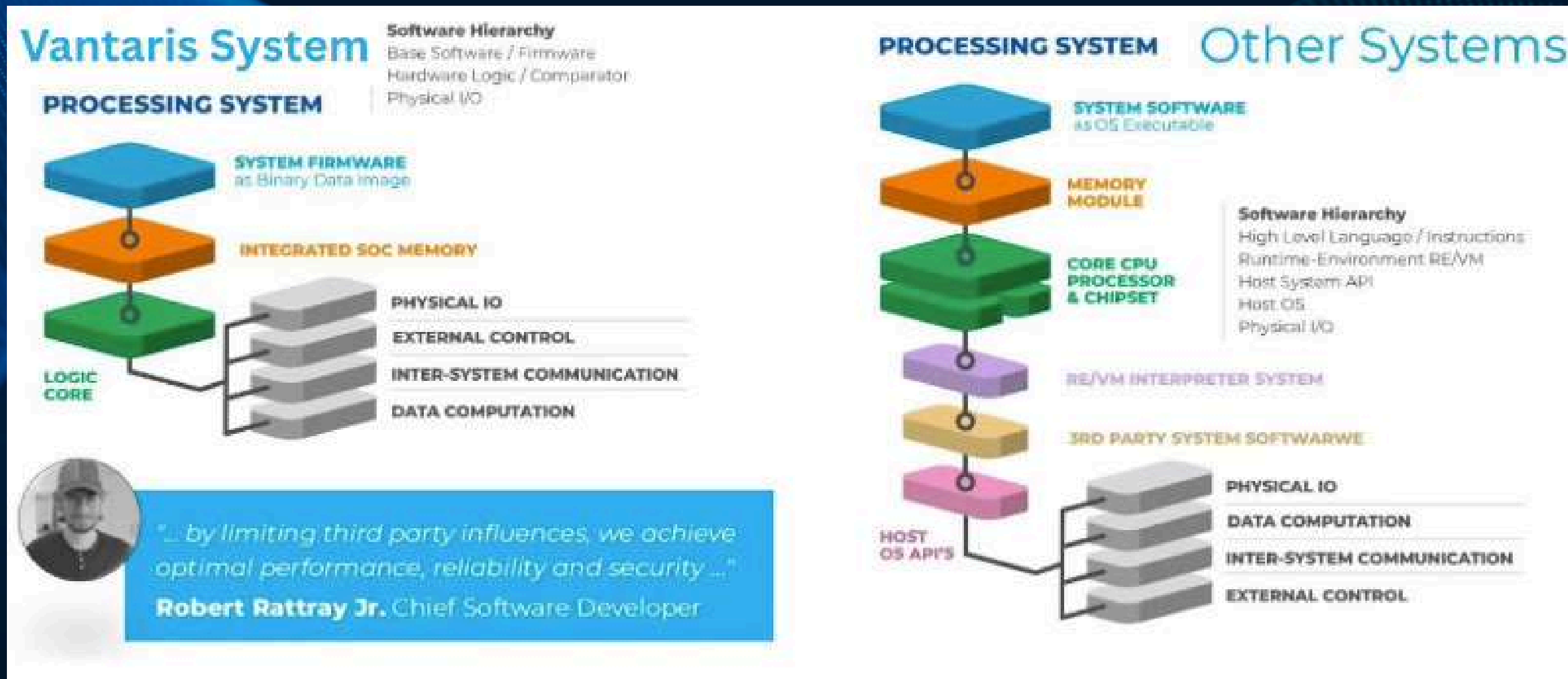
ARC is the primary long-term value and margin driver of the Vantaris platform ecosystem.

The ARC SoC integrates propulsion, navigation, weapons, optics, and communications through a single secure control architecture, eliminating latency and protocol conflicts between subsystems.

The Only sovereign-ready, unhackable command and control system for unmanned platforms.

The Autonomous Remote Command (ARC) Solution

Most deployed autonomous control systems suffer from critical structural vulnerabilities. They rely heavily on third-party commercial off-the-shelf (COTS) hardware and widely available, often open-source software stacks that must run atop general-purpose operating systems. These systems depend on multiple intermediary layers – interpreter packages (e.g., MS .NET, Java RE/VM, Python), system services (SSH/SSHD), and indirect APIs and libraries (ROS) – each introducing potential exploit vectors. Such complexity exposes platforms to security breaches, zero-day vulnerabilities, and unpredictable failures arising from incompatible or faulty subsystems – creating unacceptable risks to mission safety and operational readiness.



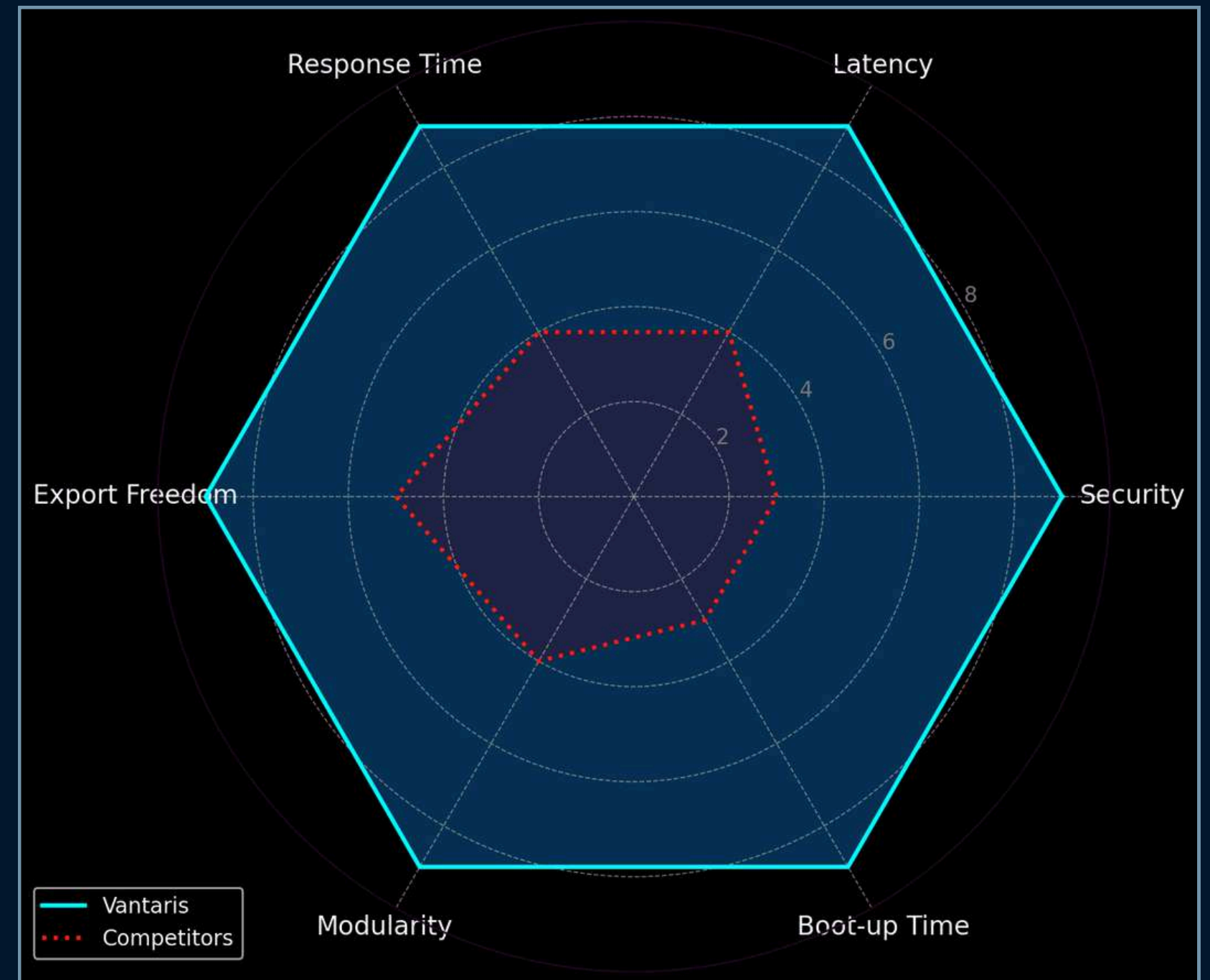
The ARC Difference

Vantaris has been purpose-built to solve the most critical challenges facing the autonomy sector today – from regulatory ambiguity to cybersecurity threats and integration rigidity.

While competitors are constrained by export controls, third-party software dependencies, and one-size-fits-all architectures, Vantaris provides a sovereign, in-country ready, security-first, and scalable autonomy platform with immediate deployment potential.

Vantaris is not just offering an autonomy product – it is delivering a strategic autonomy framework that is deployment-ready today, yet flexible enough to evolve alongside regulation and mission needs. ARC Level 3 delivers MASS Level 3 autonomy by default – and achieves Level 4 capability where conditions permit.

Once deployed, ARC becomes the vessel's non-removable command backbone.



Progressive Autonomy with Vantaris ARC Levels

The Vantaris Command and Control (C2) system offers a tiered approach to autonomous remote command (ARC), providing increasing levels of sophistication and functionality across three distinct stages.

Each level builds upon the foundational capabilities of the preceding one, ensuring a scalable and robust solution for various maritime operational requirements.

This progressive design allows for seamless integration and upgrading as operational needs evolve, culminating in highly advanced autonomous capabilities for Unmanned Surface Vessels (USVs).



ARC Level 1: Foundational Remote Control

This entry-level system establishes the core infrastructure for remote vessel operation. It provides essential control and telemetry functionalities, mirroring traditional helm operations from a remote command station. Crucially, it enables crewless operation in hazardous environments and incorporates robust safety mechanisms for critical component failures. Near real-time performance data and sensor telemetry are continuously provided to the operator, forming the bedrock for all subsequent autonomous features.



ARC Level 2: Enhanced Stability and Positioning

Building on the remote control capabilities of Level 1, ARC Level 2 introduces significant autonomous enhancements for maintaining vessel stability and position. Key features include 'Auto Helm' for fixed-course steering and 'Auto Throttle' for consistent speed management, reducing operator workload during long transits. The advanced 'Electronic Anchor' functionality allows the vessel to dynamically maintain its position within a specified radius, crucial for station-keeping and emergency situations, particularly in challenging wind and current conditions.



ARC Level 3: Advanced Mission Autonomy

ARC Level 3 represents the pinnacle of the Vantaris C2 system's autonomous capabilities, enabling sophisticated mission execution. This level integrates comprehensive waypoint navigation, allowing operators to program complex routes via a user-friendly interface. It includes critical safety features such as automatic return-to-home or last-known-comms functionality in the event of communication loss. Furthermore, Level 3 supports intelligent convoy applications, configurable system settings for dynamic mission parameters, satellite radio control for robust communication, and pre-defined manoeuvre controls for precision docking and challenging operations.



Competitive Landscape & Vantaris' Sovereign Advantage

Capability	Legacy USVs	OS-Based Autonomy	Vantaris ARC
Boot Time	3–9 mins	30–90 sec	0.1–0.2 sec
Response Latency	3–9 sec	~1 sec	100–200 ms
Architecture	Heavy OS layers	Docker/ROS/RTOS	Bare-metal (no OS)
Security	High attack surface	Medium	Minimal attack surface
Interoperability	Limited	Middleware conflicts	Native multi-domain
Sovereignty	Vendor-locked	Partially	Fully sovereign stack

Comparable NATO-exercised USVs typically cost €650k-€2m+; large OEM systems €15m-€25m+. Vantaris represents a system integrated vessel 2-20 X more cost effective.

Vantaris is the only provider offering sovereign-grade, bare-metal autonomy with ultra-low latency control and a recyclable, 3D-printed hull platform — validated through NATO partnership.

ARC is covered by proprietary firmware architecture and hardware-level sequencing that cannot be replicated using OS-based systems.

Approved Co-Development with the Portuguese Navy

Following direct presentations to the Portuguese Navy and Ministry of Defence in October 2025, Vantarix has been selected to lead a multi-phase programme delivering next-generation autonomous maritime capability.

The co-development between Vantarix and the Portuguese Navy will directly leverage Portuguese maritime expertise with advanced autonomous systems technology to collectively create operationally relevant surface vessel capabilities at competitive costs.

The Portuguese Navy will provide total and complete oversight - operational requirements and guidance, testing facilities and end user integration.

Upon successful co-development Portuguese Navy certification will be issued on completion of trials and subsequent operational use, formally recognising Vantarix as capable for NATO interoperability. The Navy will define CONOPS, supervise trials, and evaluate operational performance at every developmental stage.

In addition, Portugal will sponsor Vantarix as a Portuguese entity at REPMUS, granting direct exposure within NATO's premier unmanned systems forum.



Delivery & Integration Ecosystem

The programme unites an ecosystem of European and global partners providing complementary expertise across manufacturing, autonomy, training, and mission systems. Confirmed partners include:



CEAD

European manufacturing partner of the 8-9m multi-mission platform with extended range and interchangeable mission packets based off the same platform.

Janus Defence

CONOPS/CONEMP, training capabilities

Xiren

Manufacturing partner of the 12m multi-mission platform.

MSI

Weapon systems partner. MSI will provide weapons systems FOC to Vantar

Sea AI

Optics, (some) software for targeting & ISR

Phase 1: 8.4m Multi-Mission Platform

30 Day
Manufacture

Multi- Use

- ISR
- HVU Protection
- Kamikaze
- Contested Environments
- SWARM Capability
- FPV UAV Release

The 9 metre vessel is an ARC-3 multi-mission system with extended range and modular mission payloads.

Vantar is has partnered with CEAD, whose advanced 3D-printing capability enables rapid prototyping and precision composite production. Following validation trials, Vantar is will internalise 3D-printing manufacture, allowing scalable, in-house production of operational platforms.

The vessel has been fully designed, tested, and is production-ready, forming the foundation for initial deployment under the Portuguese Navy programme.



Phase 1: Vessel 1 Build Cost

Verified Supplier Mid-Point Values (EUR)

Cost Component	USD (Mid-Point)	EUR (Mid-Point)
Design Fee (NRE – one-off)	\$17,500	€15,750
HDPE+GF 3D-Printed Hull	\$58,500	€52,650
Hull Outfitting (systems & electrics)	\$12,500	€11,250
Engine + Bravo III Drive + Steering	\$50,500	€45,450
Subtotal – Supplier Cost Basis	\$139,000	€125,100
Versatile Marine Margin (35%)	\$42,500	€39,800
Total Build Cost (Supplier Mid-Points incl. Margin)	\$181,500	€164,900

Additional Delivery & Commissioning Costs for vessel 1: €14,000

- **Vessel #1: €178,900**
- **Vessel #2 onward (CEAD): €163,150**
- **Further ~10% reduction achievable at series scale**
- **Additional ~15% reduction achievable under future in-house manufacture (excluding amortised CapEx)**

This includes all supplier costs, builder margin, shipping, and commissioning preparation
Excludes Vantaris' system-integrator margin
NRE is a one-time cost for the first vessel only

Steady-State Reference Case (Used for Forward Modelling)

- **CEAD-based vessel build cost: ~€130,000**
- **Vessel gross margin: ~€85,000 (~40%)**

System Sale Price: €215,000

- **System Margin: €85,000**

System Sale Price (Inc ARC):

€320,000 - €390,000

- **System Margin (Inc ARC): €169,000-€225,000 (52%-57%)**

*****Margins improve materially as ARC and TLS scale across fleet deployments**

ARC Hardware Pricing (Excluding Payloads)

No licence fees. No Subscriptions. Payloads are customer-selected

ARC pricing scales linearly with every vessel deployed and is independent of hull supplier

ARC 1: €105,000

ARC 2: €140,000

ARC 3: €175,000

Vantaris does not supply mission payloads (optics, EO/IR, radars, weapons, sonars). We ensure seamless integration with any customer-selected optics, radar, EW, sonar, or weapon systems through ARC's open, hardware-level protocol architecture.

ARC Margin: 80%+



Through-Life Support (TLS)

Annual Fee: €32k - €39k (10%) per vessel

Margin: 55% - 60%

Contract Term: 5 years

Preventative Maintenance Support

- Scheduled maintenance plans (engine, driveline, electrics, ARC enclosure)
- Checklists, documentation & compliance records

Corrective Maintenance & repairs

- Remote diagnostics & engineering support
- Access to accredited repair partners
- Guaranteed availability of all required parts
(replacement parts billed separately – standard naval model)

ARC Firmware & System Servicing

- Firmware updates
- Cyber/patching
- System health monitoring
- Mission-system compatibility updates

Field Engineering Support

- Remote support included
- On-site support available (day rate)

Training Packages (Operators & Maintainers)

Delivered by Janus Defence

Operator Training (per cohort of 10): €30,000 - €40,000

Maintainer Training (per cohort of 10): €40,000 - €55,000

Optional:

- Annual Recertification: €8,000
- On-Site Support: €2,000 - €3,000 / day

Training Margin:

80%



Unit Economics Overview (Phase 1 - Series Production)

Validation-stage economics prior to fleet-scale deployment

1. Vessel Margins				
Vessel Stage	Customer Price	Internal Cost	Margin (€)	Margin %
Vessel #1 (with NRE)	€215,000	€178,900	€36,100	17%
Vessel #2+ (no NRE)	€215,000	€163,150	€51,850	24%
Series Production Vessel	€215,000	€130,000	€85,000	40%

2. ARC Hardware Margins (80%)				
ARC Level	Customer Price	Internal Cost (20%)	Margin (€)	Margin %
ARC 1	€105,000	€21,000	€84,000	80%
ARC 2	€140,000	€28,000	€112,000	80%
ARC 3	€175,000	€35,000	€140,000	80%

3. Through-Life Support (TLS) Margin (55%)				
Annual service	Customer Price	Internal Cost (45%)	Margin (€)	Margin %
TLS (per vessel, per year)	€32,000 – €39,000	€14,400 – €17,550	€17,600 – €21,450	55%

4. Training & Certification Margins (80%)				
Operator Training				
Training Cohort	Customer Price	Internal Cost (20%)	Margin (€)	Margin %
Operator Training (10 personnel)	€35,000	€7,000	€28,000	80%
Maintainer Training				
Training Cohort	Customer Price	Internal Cost (20%)	Margin (€)	Margin %
Maintainer Training (10 personnel)	€50,000	€10,000	€40,000	80%

Phase 1 8.4m USV - 10 Vessel Contract

Financials

This table illustrates the unit economics of a representative Phase 1 contract for ten 8.4m USVs.

The model reflects Vantaris’ component-level commercial structure, with the platform, ARC Level 2 autonomy, Technical Lifecycle Support (TLS), and training contracted separately.

Platform pricing and costs are based on validated CEAD manufacturing assumptions.

TLS is priced at 10% of system value per vessel per year over five years, and training assumes a single operator and maintainer cohort (10 personnel each). The example is intentionally conservative and excludes payloads, upgrades, additional training, integration work, and fleet expansion.

Phase 1 – 8.4m USV 10-Vessel Programme Economics (5-Year View)				
<i>(Component-level commercial model: platform and ARC sold separately; ARC 2 assumed.)</i>				
Revenue Stream	Revenue (€)	Cost (€)	Gross Margin (€)	Margin %
Platform (10 x €215k)	€2,150,000	€1,300,000	€850,000	39.5%
ARC 2 Autonomy (10 x €140k)	€1,400,000	€280,000	€1,120,000	80%
Technical Lifecycle Support (10 vessels x 5 yrs)	€1,600,000	€720,000	€880,000	55%
Operator Training (10 personnel)	€35,000	€7,000	€28,000	80%
Maintainer Training (10 personnel)	€50,000	€10,000	€40,000	80%
Total Programme	€5,235,000	€2,317,000	€2,918,000	55.8%

Phase 2: 12-30m Multi-Mission Platform

Phase 2 platform expansion post-REPMUS; not required for Phase 1 validation

Xiren is a next-generation marine technology company specialising in advanced hull-form design and high-performance vessel construction. Its flagship military platform, the X12, is a 12-metre, high-speed multi-role craft engineered to exceed the operational requirements of modern defence and security forces.

The X12 delivers exceptional performance across speed, agility, and payload capacity. Its proprietary hull form provides unmatched stability and control at speeds exceeding 60 knots, while maintaining ultra-low drag and superior sea-keeping in heavy conditions. The design enables rapid acceleration, reduced fuel consumption, and minimal operator fatigue – key advantages for both manned and unmanned deployment.

Built with military-grade aluminium & composite structures and configurable internal layouts, the X12 supports a range of mission profiles, from high-speed interdiction and surveillance to autonomous coastal defence. When paired with Vantar's Command and Control system, the platform becomes a fully integrated, optionally-manned solution – capable of seamless transition between human and remote operation with sub-200ms command latency.



Transition to In-House Manufacturing (LSAM / CEAD)

Large-Scale Additive Manufacturing (LSAM) – Phase 2+ Capability)

12m LSAM

- Prints recyclable HDPRO hulls and structural components
- 20-100 kg/hr extrusion rate
- Hull sections or full hulls printed in under 6 days
- Suitable for rapid development of 8m-12m platforms

Capital Profile

Acquisition: €750k-€1m

Lease option: €8k-€15k / month

Strategic Role (Not Mass Production)

LSAM is deployed as a development, acceleration, and sovereignty tool, not as a sole volume-production method.

Primary functions:

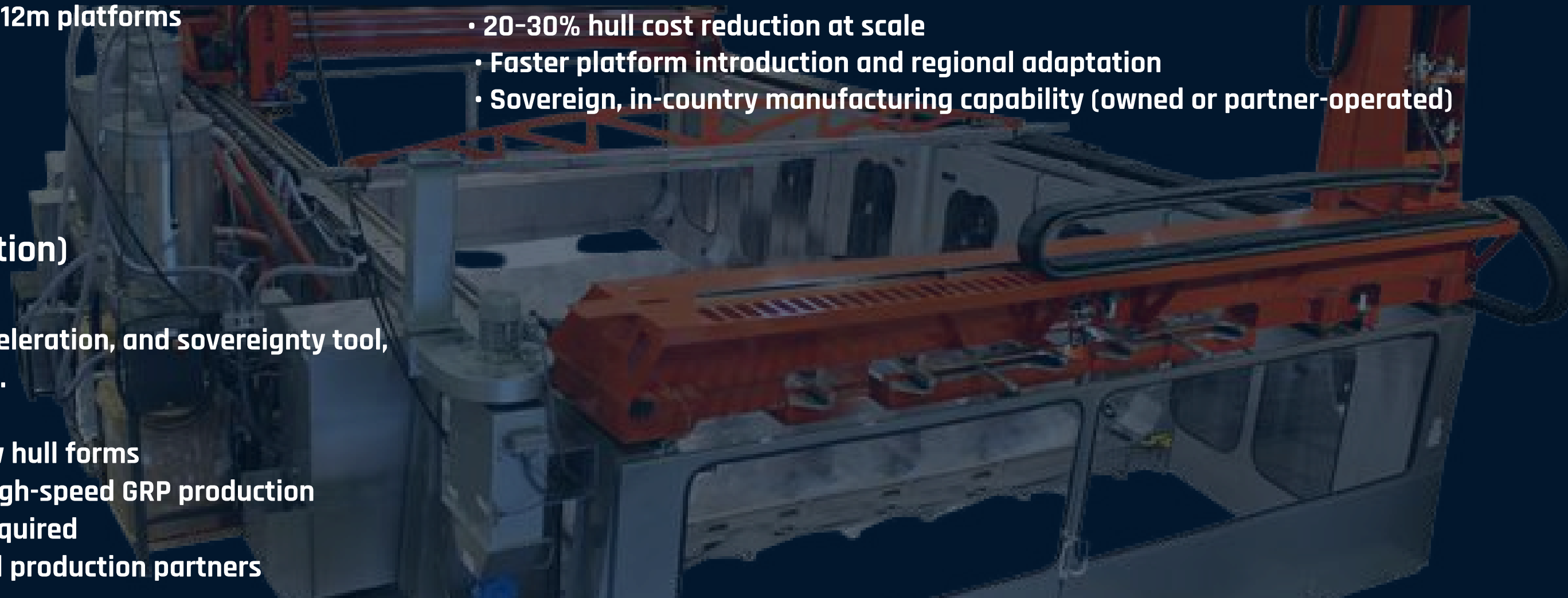
- Rapid prototyping and iteration of new hull forms
- Plug & mould manufacture enabling high-speed GRP production
- Low-volume sovereign builds where required
- Validation and localisation for regional production partners

Scale & Expansion Model

- Fleet-scale production achieved through hybrid manufacturing
- GRP hulls produced in parallel via established regional partners (e.g. CEAD, international yards)
- LSAM enables rapid platform rollout; partners deliver throughput

Impact (Phase 2 & Beyond)

- 20-30% hull cost reduction at scale
- Faster platform introduction and regional adaptation
- Sovereign, in-country manufacturing capability (owned or partner-operated)



Phase 1 Delivery: Path to REPMUS Validation & Required 12-Month Budget

This round funds validation and NATO-facing demonstration; commercial scale follows certification.

Category	12-Month Cost (€)
Vessel Build (8.4m Phase 1 Platform)	€179,000
Personnel (5 FTEs)	€330,000
Office + Industrial Space	€55,000
Travel, Deployment & Operations	€80,000
Legal, Accounting & Compliance	€30,000
Contingency	€76,000
 Total Requirement	€750,000

Phase 1 is focused on delivering and validating a mission-ready autonomous maritime capability through Vantar's formal co-development programme with the Portuguese Navy and Ministry of Defence. The objective is to demonstrate advanced autonomy, multi-mission capability and NATO interoperability at REPMUS 2026, NATO's flagship exercise for uncrewed and autonomous systems.

At this stage, the priority is execution rather than speculative forecasting. Success will establish operational credibility, generate real-world performance data and validate the technology alongside end-users, creating a foundation for future procurement, strategic partnerships and programme expansion.

For investors, this represents a highly differentiated entry point: participation alongside a government-backed development programme, a clear pathway to NATO validation and exposure to one of defence's fastest-growing sectors.

Defence Adoption Pathway & Go-To-Market Strategy

01

1. Phase 1 Co-Development (Now)

- Build, integrate, and trial the first platforms with the Portuguese Navy
- ARC validation across mission profiles
- Joint testing & Navy acceptance

02

2. REPMUS 2026 (Catalyst Event)

- Live NATO demonstration
- Real-world data capture and evaluation
- Exposure to procurement teams from 20+ NATO nations
- Platform becomes “validated” in a Tier-1 environment

03

3. Post-REPMUS Acceleration (Months Following)

- Bilateral demonstrations to interested NATO partners
- Follow-on opportunities with Portugal
- Expansion to other NATO partners
- Engagement with US-aligned procurement pathways (NSPA, bilateral programmes)

04

4. Phase 2 (Series A) - Scaling Production

- In-house LSAM manufacturing (12m printer)
- Expanded vessel family (8m → 12m)
- Expansion into UGVs & possible UAVs
- Lower cost base, 35-40% vessel margins
- Fleet-level ARC deployments & recurring TLS/CEP revenues

Global Shift Toward Autonomy

The past decade has seen a decisive global shift toward autonomous and optionally manned systems, particularly in defence and maritime sectors. As operational demands grow and cost-efficiency becomes critical, autonomy is moving from experimental to essential.

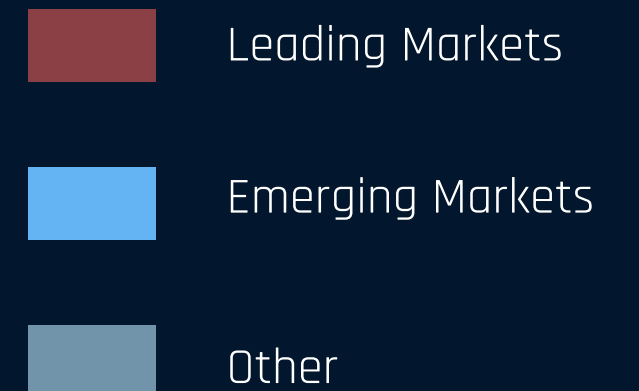
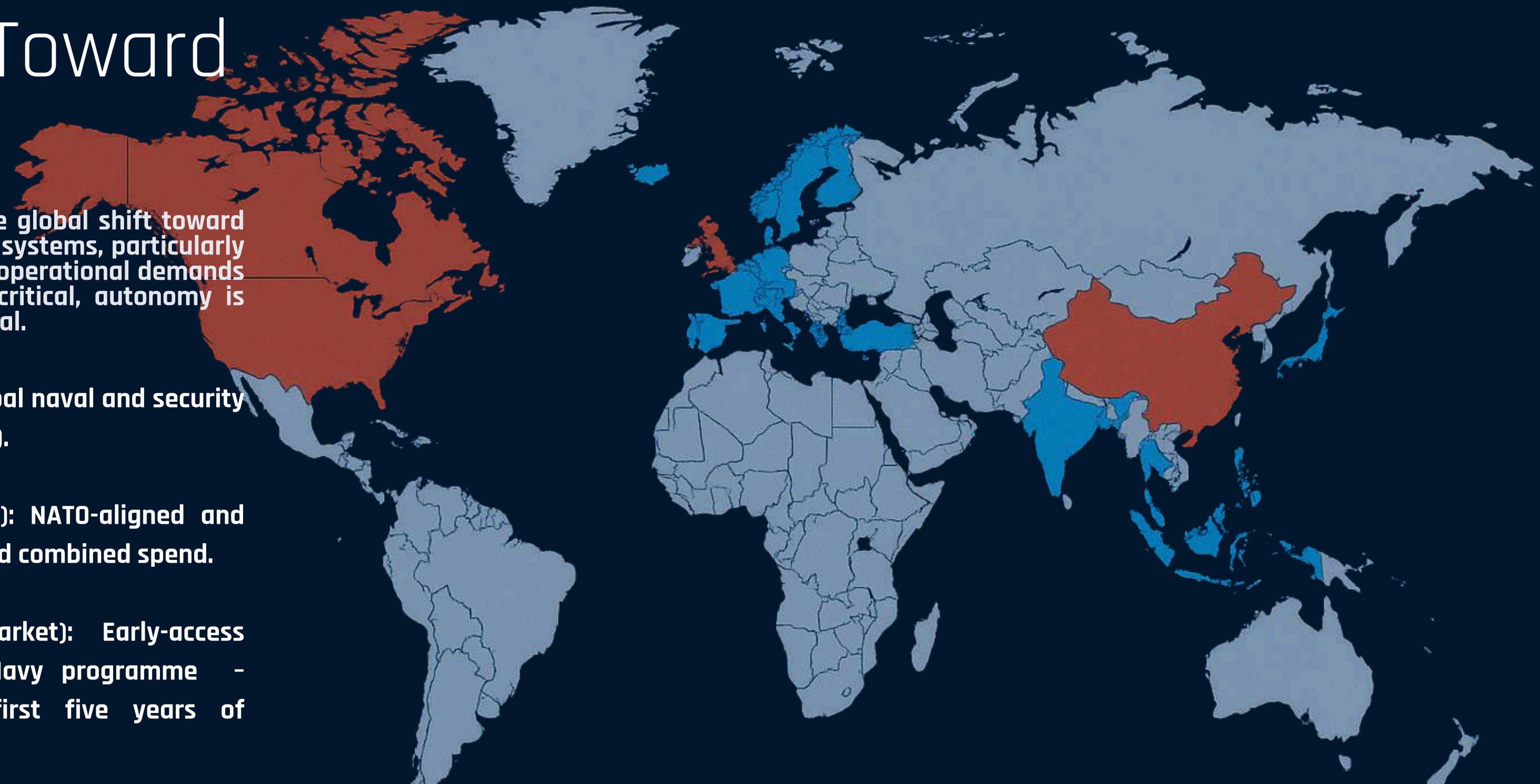
TAM (Total Addressable Market): Global naval and security USV demand (NATO, GCC, Asia-Pacific).

SAM (Serviceable Available Market): NATO-aligned and GCC adoption zones - \$3.5B estimated combined spend.

SOM (Serviceable Obtainable Market): Early-access opportunity through Portuguese Navy programme - \$300-500M addressable within first five years of production.

Vantaris' validated technology and low-cost production model position it uniquely within the most active and highest-barrier segments of the autonomy market.

Early adoption driven by defence procurement, not commercial forecasting assumptions.



Why Invest Now – The vantar is Inflection Point

A low-risk entry point into a high-growth, sovereign autonomy company aligned with NATO demand and global defence trends.

This round materially de-risks the business ahead of Series A by delivering NATO-recognised validation.

- Backed by the Portuguese Navy under a formal co-development programme
- Guaranteed platform showcase at REPMUS 2026, NATO's premier autonomy exercise
- First NATO-ready USV with a printed hull and sovereign bare-metal autonomy stack
- Multi-layer margin model: ARC (80%), TLS (55%), CEP (70%), Training (50%)
- Phase 1 is execution-only with clear deliverables and low uncertainty
- €750k raise fully funds Phase 1
- Post-REPMUS 2026 expected procurement interest & commercial momentum
- Phase 2 (Series A) unlocks in-house LSAM manufacturing and 35-40% vessel margins
- Positioned for NATO + EU + GCC procurement pathways



www.vantaris.io

jack@vantaris.io

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